NMC 2013 PRIMARY LEVEL PAPER 1 QUESTIONS

SECTION A : 5 marks each

Qn. 1. What is half of 1.01?

Qn. 2. At a Physical Education session, 30 children were made to stand in each column and 16 columns were formed. If 24 children had been made to stand in each column, how many columns could have been formed?

Qn. 3. Insert either a + or a − in either of the boxes so that the result of the calculation is correct: $123 \square 45 \square 67 \square 89 = 100$.

Qn. 4. Divide 20132013 by 2013.

Qn. 5. The operation $\oplus$ is defined as $a \oplus b = a^2 + a + b$. What is the value of $-5 \oplus 4$?

Qn. 6. Ruth looks at the word JUNIOR in a mirror. Which reflected letters look the same, no matter how Ruth holds the mirror and why?

Qn. 7. A man ate 100 pancakes in five days, each day eating 6 more than the previous day. How many pancakes did he eat on the second day?

Qn. 8. If your watch loses 5 minutes each hour and you set the time correctly at 7 : 00. What is the actual time when your watch shows later that morning that it is 09 : 45?

Qn. 9. Peter buys a pizza and eats half of it on the first day. On the second day, he eats one-third of the remaining part. what fraction of the original pizza is still uneaten?

Qn. 10. IF IT IS BLUE, IT IS ROUND
IF IT IS SQUARE, IT IS RED
IT IS EITHER BLUE OR YELLOW
IF IT IS YELLOW, IT IS SQUARE
IT IS EITHER SQUARE OR ROUND.
If all the above statements are true, we can conclude that (*circle the correct answer*):
A) It is red *B*) It is a blue square *C*) It is red and round
D) It is yellow and round *E*) It is blue and round

SECTION B : 10 marks each

Qn. 11. Brenda wants to complete the diagram below by inserting three numbers, one into each empty cell. She wants the sum of the first three numbers to be 100, the sum of the middle three numbers to be 200, and the sum of the last three numbers to be 300. What number should Brenda insert into each of the empty cells?

10	?	?	?	130

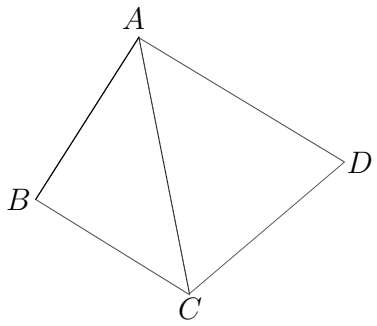
Qn. 12. The rectangle shown in the diagram is divided into 7 squares. One big square marked A, three similar smaller squares are marked B and another 3 smallest similar squares marked C. The area of each of the 3 squares marked B is 64cm^2.

(a) What is the area of square marked A?

(b) Calculate the perimeter of the rectangle.

Qn. 13. Jane is making patterns using identical matchsticks. Jane needs 4 matchsticks to make a 1×1 pattern; 12 matchsticks to make a 2×2 pattern, 24 matchsticks to make a 3×3 pattern and 40 matchsticks to make a 4×4 pattern. How many matchsticks does Jane need to make a 15×15 pattern?

Qn. 14. The diagram shows a quadrilateral $ABCD$ in which $AD = BC$, angle $CAD = 50^o$, angle $ACD = 65^o$ and angle $ACB = 70^o$. What is the size of angle ABC?



Qn. 15. One packet of sweets costs Shs. 1000. There is a promotional voucher inside every packet. For every three promotional vouchers you collect, you get a free packet of sweets. How many packets of sweets will you get for Shs. $15,000$?

NMC 2013 PRIMARY LEVEL PAPER 1 SOLUTIONS

PRIMARY PAPER 1 SOLUTIONS

SECTION A

Qn. 1. $\frac{1}{2} \times 1.01 = 0.505$.

Qn. 2. The total number of children is $30 \times 16 = 480$ children. The number of columns are $\frac{480}{24} = 20$.

Qn. 3. $123 - 45 - 67 + 89 = 100$.

Qn. 4. $20132013 = 2013000 + 2013 = 2013 \times 1000 + 2013$.
Therefore $\frac{20132013}{2013} = \frac{2013 \times 1000}{2013} + \frac{2013}{2013} = 1000 + 1 = 1001$.

Qn. 5. $-5 \oplus 4 = (-5)^2 - 5 + 4 = 25 - 5 + 4 = 24$.

Qn. 6. The letters are U,I and O.
Reason. These are the only letters that have got lines of symmetry.

Qn. 7. Suppose the man ate p pancakes on the first day, then he ate $p + 6$, $p + 12$, $p + 18$, $p + 24$ respectively on the second, third, fourth and the fifth day.
$p + (p + 6) + (p + 12) + (p + 18) + (p + 24) = 100 \Longrightarrow 5p + 60 = 100$. Hence $p = 8$.
On the second day, he ate $8 + 6 = 14$ pancakes.

Qn. 8. If the watch loses 5 min every hour, then the time on the watch increases by 55 minutes for every hour that passes.
On the watch, from $07 : 00$ to $09 : 45$ is $2 \times 60 + 45 = 165 = 3 \times 55$. The true time that has passed is 3×60 minutes or 3 hours.
True time is $10 : 00$.

Qn. 9. $\frac{1}{2}$ of the pizza is left after the first day. On the second day, he eats $\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$. Therefore what remains of the original pizza is $\frac{1}{2} - \frac{1}{6} = \frac{1}{3}$.

Qn. 10. From the third statement, it is either yellow or blue but from the fourth and second statements, if it is yellow, then it is red which is a contradiction. Therefore it is blue. Also from the first statement, it is also round.

SECTION B : 10 marks each

Qn. 11.

10	x	y	z	130

$$10 + x + y = 100 \quad (4)$$
$$x + y + z = 200 \quad (5)$$
$$y + z + 130 = 300 \quad (6)$$

From equation (3), $y + z = 170$ and substituting this in equation (2) gives $x + 170 = 200 \Longrightarrow x = 30$.
$10 + x + y = 100 \Longrightarrow 10 + 30 + y = 100$ and so $y = 60$.
$x + y + z = 200 \Longrightarrow 90 + z = 200$ and so $z = 110$. We thus have

10	30	60	110	130

Qn. 12. (a) Square A is $3x$ by $3x$cm and square C is x by x cm and square B is 8 by $8cm$
This implies that $4x = 8 + 8 + 8 = 24 \Longrightarrow x = 6$
Area of A is $18 \times 18 = 324cm^2$

(b) Perimeter of the rectangle is $= 24 + 24 + 26 + 26 = 100$cm

Qn. 13. The sequence $4, 12, 24, 40, \cdots$ can also be written in terms of the size of the pattern made as

$$1 \times 4, \quad 2 \times 6, \quad 3 \times 8, \quad 4 \times 10, \quad \cdots, \quad n \times (2n + 2)$$

If a 15×15 pattern is required, it means that $n = 15$.
The matchsticks required are $15 \times (2(15) + 2) = 15 \times 32 = 480$.

Qn. 14. Angle $ABC = 50^o$. ACD is an isosceles triangle $\overline{AC} = \overline{AD} = \overline{BC}$. Triangle ABC is also an isosceles triangle . Therefore $< BAC =< ABC = 55^o$.

Qn. 15. 15000 buys 15 packets including 15 promotional vouchers.
15 promotional vouchers give 5 more packets(and 5 more promotional vouchers).
5 promotional vouchers give 1 more packet and 1 more vouchers.
3 promotional vouchers give 1 more packets and 1 more unusable vouchers.
Therefore the total number of pockets is $15 + 5 + 1 + 1 = 22$ packets.